

Priyanshi Jain

New York, USA

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Education

Stony Brook University

Master's in Data Science

– **Relevant Coursework:** Data Analysis, Probability & Statistics, NLP, LLMs, Statistical Learning

Stony Brook, NY

Aug 2025 - May 2027

SVKM's NMIMS

Bachelor of Technology in Computer Engineering

– **Relevant Coursework:** AI & ML, Deep Learning, Natural Language Processing, Reinforcement Learning, Data Mining

India

Aug 2021 - May, 2025

Technical Skills

Languages: Python, SQL, R, MATLAB, SAS, Java, C++, JavaScript, TypeScript

AI/ML & NLP: Scikit-Learn, NumPy, Matplotlib, TensorFlow, PyTorch, RASA, LLaMA 3, vLLM, LightGBM, CatBoost, FinBERT, HuggingFace Transformers, spaCy, Time Series Modeling, Conversational AI Systems, Feature Engineering, Statistical Hypothesis Testing

Analytics & BI Tools: Tableau, Excel, PowerBI, Plotly, Streamlit, Jupyter Notebook, EDA, ETL

Data: PostgreSQL (Query Optimization, Data Modeling), Pandas, MongoDB, Django, REST APIs, JSON/CSV

Cloud & DevOps: AWS, Azure, Vercel, Git, Docker, Kubernetes, CI/CD, GCP, Hadoop, Airflow, Bash, Grafana

Experience

Software Engineering Intern – *Answering Legal*

Apr 2026 – Present

- Built and scaled a **production-grade conversational AI platform** using **React, Django REST APIs, and Rasa**, supporting multiple call flows, dynamic routing, and real-time user interactions
- Designed and implemented **custom LLM training pipelines** by synthesizing domain-specific datasets and fine-tuning models (**PyTorch, vLLM**), improving response accuracy and reducing hallucinations in high-frequency use cases
- Developed **admin-facing configuration systems** enabling dynamic control of bot behavior (greetings, FAQs, call routing), and engineered **monitoring pipelines** to track latency, performance metrics, and lead-generation outcomes
- Optimized backend performance through **Dockerized microservices** and **PostgreSQL query tuning**, reducing API latency and enabling scalable, high-throughput conversational workloads

AI/CV Intern – *iKites Technologies Pvt. Ltd.*

Mar 2025 - Aug 2025

- Conducted large-scale data evaluation for Deepfake Detection models across 5,000+ insurance claim images, benchmarking SightEngine and Gemini APIs, built evaluation frameworks, metric computation (Precision/Recall/F1), improving evaluation speed by **80%**
- Formulated **data processing pipelines** for SD-OCT retinal imaging, **performing segmentation, annotation, and quantitative analysis** over **1,200+** scans, supporting model development that achieved **94% accuracy** on retinal scan classification
- Developed **analytical dashboards** to **visualize model metrics, failure patterns**, and comparative performance trends, supporting **data-driven decision-making** for stakeholders

Software Development Intern – *Beyond Key*

June 2024 - July 2024

- Designed frontend modules in **Angular** for Trackr, a dental tracking application; improved component reusability by **30%**
- Built RESTful APIs and backend architecture with **Node.js & MongoDB** for real-time clinic workflow management
- Practiced **Agile** methodologies and collaborated with **QA** for continuous integration and deployment

Projects

Earnings Call Intelligence Engine 🌀 | *Python, FinBERT, spaCy, Loughran-McDonald, Streamlit*

- Crafted an NLP pipeline processing **137 SEC earnings call transcripts** across **16 SP 500** companies, **scoring 68,909 sentences** using **FinBERT** and **Loughran-McDonald dictionary** across 7 linguistic dimensions, with section-level parsing (prepared remarks vs. Q&A), speaker role classification, and forward-looking statement detection via **spaCy PhraseMatcher**
- Identified that uncertainty language in scripted remarks is the strongest short-term return predictor (**Spearman = 0.166, p = 0.053**) through rigorous statistical analysis including **Bonferroni-corrected hypothesis testing**, divergence quartile analysis, and a quarter-over-quarter language shift detector that flagged verifiable corporate events like Target's 2022 inventory crisis

HDC-RAG: LLM Hallucination Analysis Framework | *Python, PyTorch, RAG, HuggingFace, Transformers*

- Evaluated hallucination rates across **4 open-source LLMs** (**Phi-4-mini, Mistral-7B, Qwen2.5-7B, Llama-3.1-8B**) on **HotpotQA** and **TruthfulQA**, engineering a 24-way evaluation matrix (4 models × 3 prompt variants × 2 datasets) to quantify prompt sensitivity
- Established a **6-class hallucination taxonomy** using **DeBERTa-MNLI** cross-encoder for NLI-based classification with per-class precision tracking; implemented inference via **Ollama/HuggingFace** backends with 4/8-bit quantization for multi-model comparison

Time Series Forecasting Competition | *Python, LightGBM, Feature Engineering, Time Series*

- Constructed a leakage-safe pipeline for **hierarchical time series** across multiple groups (code, sub-code, sub-category, horizon) and curated 50+ features including lag, rolling statistics, difference, and category-level aggregations, **ranked 118/1500 (top 10%)**
- Trained **gradient boosting models** (**CatBoost, LightGBM**) with time-aware validation, ensured strict sequential prediction to prevent look-ahead bias. Improved prediction robustness via **model ensembling** and horizon-specific modeling

Research work and publications 🌀

An Analytical Review of Social Media-Based Sentiment Analysis Techniques

- Published in the international Springer conference: Modern Practices and Trends in Expert Applications and Security (MP-TEAS)

Certifications and Courses

- **Machine Learning Specialization** - Andrew Ng (deeplearning.ai)
- **Deep Learning Specialization** — Andrew Ng (deeplearning.ai)
- **RPA Associate** — UiPath
- **Agile Methodology Virtual Experience** — Cognizant